

Topic 2



Lecture 2 · Exploratory Data Analysis (EDA) Part 3

National University of Singapore (NUS)

TODAY'S OBJECTIVES

1

Association

2

Response and Exploratory Variables

3

Association Between 2 Variables

- 2 Categorical: Contingency Table, Bar Plot
- 1 Quantitative and 1 Categorical: Box Plot
- 2 Quantitative: Scatter Plot, Correlation Coefficient

01

Association

What is an Association?



DEFINITION

An association exists between two variables if a particular value for is more likely to occur together with certain values of the other variable.



75 – 85 kg

More likely

45 – 55 kg

Less likely

1.8 - 1.9 m



KEY INSIGHT

Associations are explored using both numerical and graphical summaries

02

Response vs Explanatory Variables

Response vs. Explanatory Variables

EXPLANATORY VARIABLE

- The variable believed to influence the response variable.
- Also known as: **independent variable, covariate, or regressor**
- When categorical, it defines the groups that will be compared.

EXAMPLE

Smoking Status (Yes/No)

RESPONSE VARIABLE

- The variable on which comparisons are made
- Also known as: **dependent variable**

EXAMPLE

Lung Cancer Status (Yes/No)

Note: In some studies, it may not be possible to clearly assign roles. In such cases, we simply explore the association without assigning specific roles.

Example: Breast Cancer Study Dataset

Study: In 1900, 1,200 post-menopausal women followed for 10 years to investigate whether Post-Menopausal Hormone (PMH) therapy was associated with breast cancer incidence.

Variable	Description	Type	Coding
Age	Age at start of study (1990)	Quantitative Continuous	—
PMH	Use of post-menopausal hormone therapy	Categorical Nominal	2 = No, 3 = Yes
BBD	Developed benign breast disease	Categorical	Yes / No
BMI	Body Mass Index in 1990	Quantitative Continuous	—

Objective: explore the association between the response variable (BBD) and each explanatory variable separately.

03

Association Between 2 Variables

Two Categorical Variables

1

Contingency Table

2

Bar Plot

Two Categorical Variables: Contingency Tables

CONTINGENCY TABLE (pmh.use × Cancer)

pmh.use	Cancer	
	Absent	Present
No	682	318
Yes	121	79

Cancer	pmh.use	
	No	Yes
Absent	682	121
Present	318	79

- Tabular display to summarise the relationship between 2 categorical variables
- Rows represent categories of one variable. Columns represent categories of the other
- More commonly: response variable (Cancer) is placed in the columns. Explanatory variable (PMH use) in the rows.

Column-Wise Percentage Table

To investigate association, convert frequencies to percentages within each explanatory variable group:

Cancer	pmh.use	
	No	Yes
Absent	68.2	60.5
Present	31.8	39.5

MEASURES OF ASSOCIATION

DIFFERENCE IN PROPORTIONS

$$39.5\% - 31.8\% = 7.7 \text{ percentage points}$$

A larger difference indicates a stronger association.

RELATIVE RISK (RR)

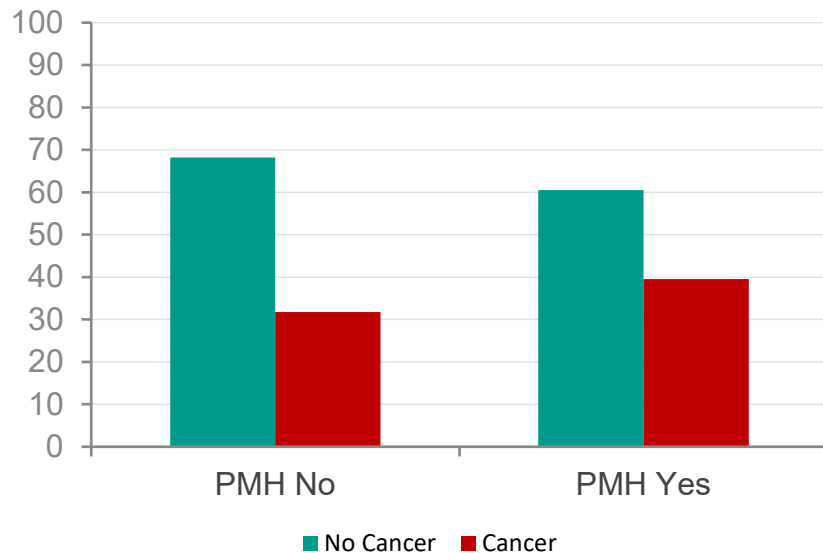
$$\frac{39.5}{31.8} = 1.24 \quad \text{PMH users have 1.24x the probability of cancer}$$

If RR differs substantially from 1, association is suggested.

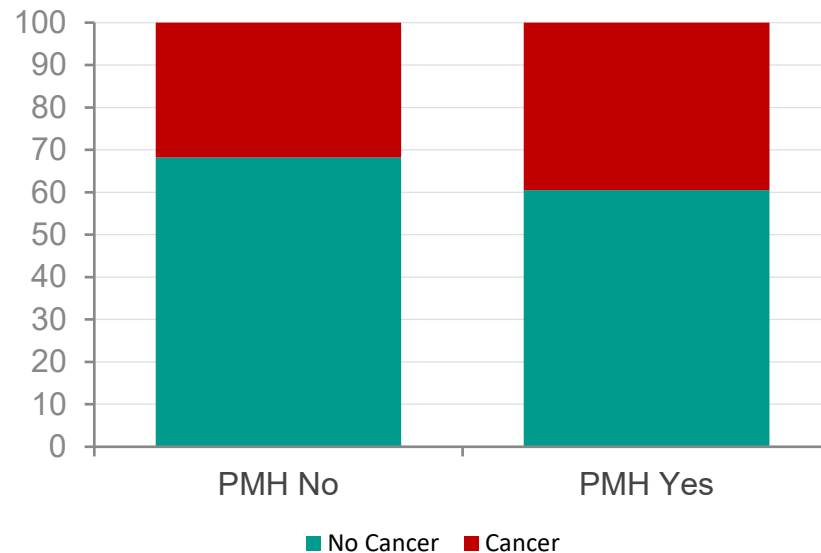
Visualising Categorical Associations: Bar Plots

Two common graphical summaries for contingency tables:

Clustered Bar Plot



Stacked Bar Plot



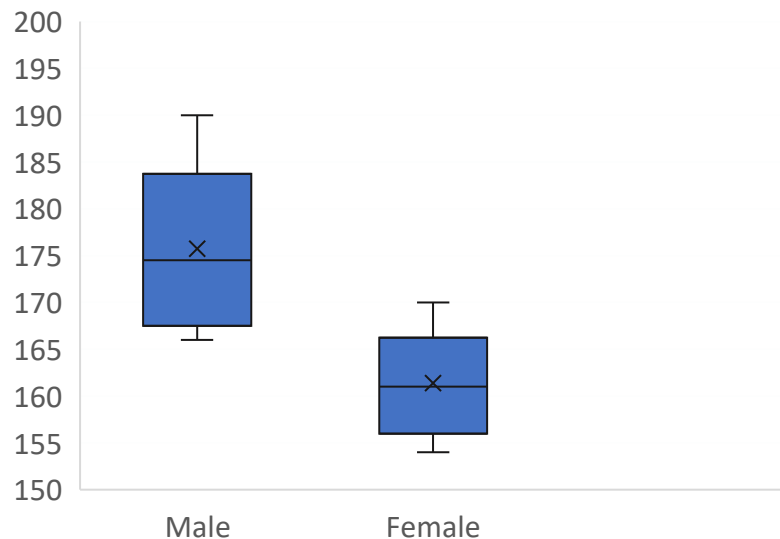
1 Quantitative & 1 Categorical Variable

1

Box Plot

Comparing Two Populations With Boxplots

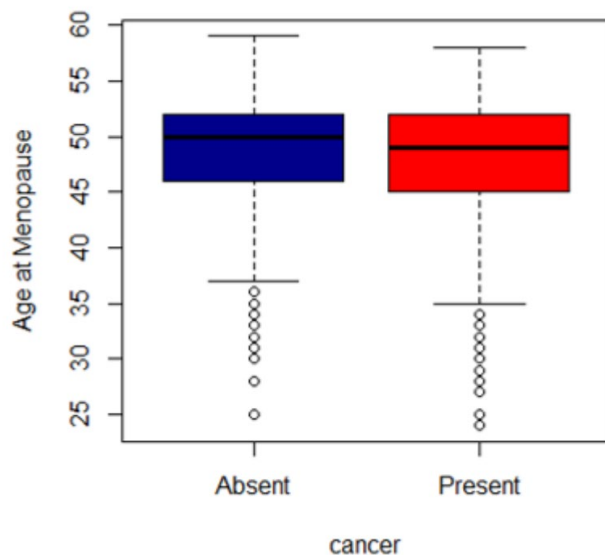
Method: divide the data into groups based on the categorical variable, then compare the distributions of the quantitative variable using boxplots placed side by side.



- Male boxplot appears higher than the female boxplot
- Males tend to have taller heights than females
- Possible association between gender and height

Comparing Two Populations With Boxplots

Method: divide the data into groups based on the categorical variable, then compare the distributions of the quantitative variable using boxplots placed side by side.



- Is age at menopause is associated with the incidence of breast cancer?
- From the plot:
 - Both groups contain many outliers below median
 - Left-skewed: many women experienced menopause at relatively young ages
 - The spreads of the two groups are similar

Conclude: no strong evidence for association between age and cancer

Two Quantitative Variables

1

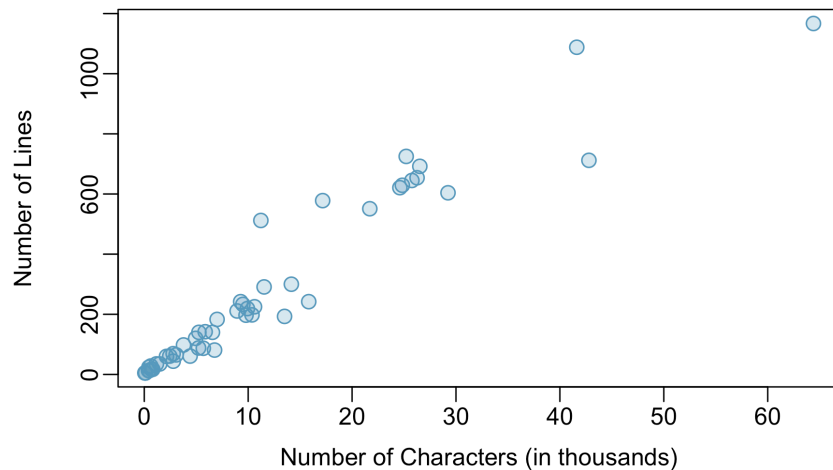
Scatter Plot

2

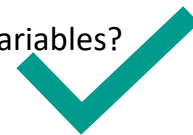
Correlation Coefficient

Two Quantitative Variables: Scatter Plots

Each observation is a point on the plot. 50 emails = 50 points



Is there an association between the 2 variables?



Number of
Line Breaks

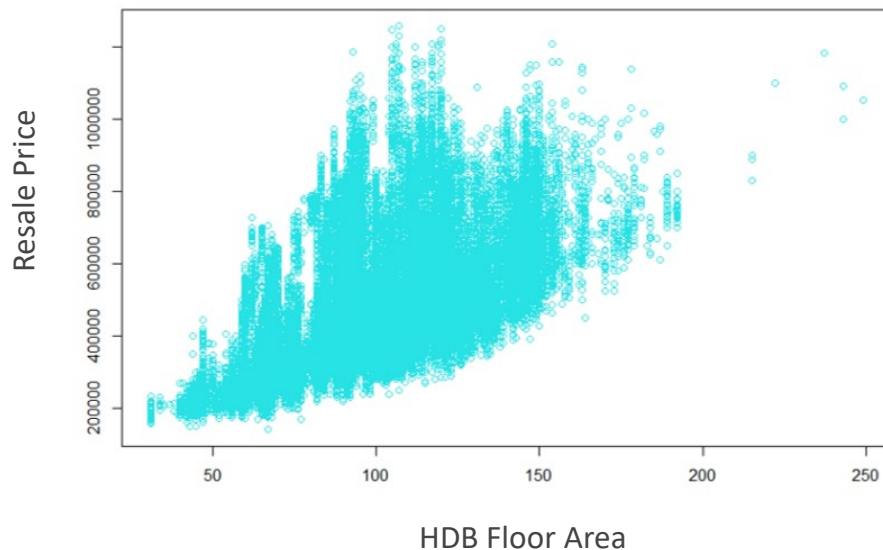


Number of
Characters

Positive Linear Association

Two Quantitative Variables: Scatter Plots

99119 HDB resale transactions in Singapore between January 2017 and June 2021



- The response variable, Resale Price, is plotted on the y-axis

Is there an association between the 2 variables?

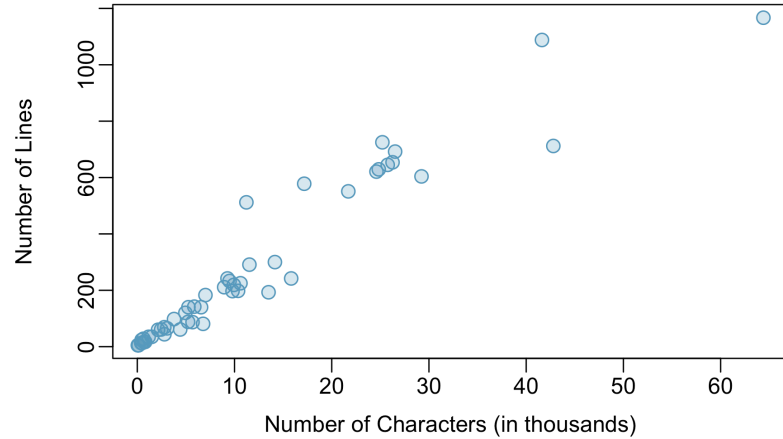


↑
Floor
Area

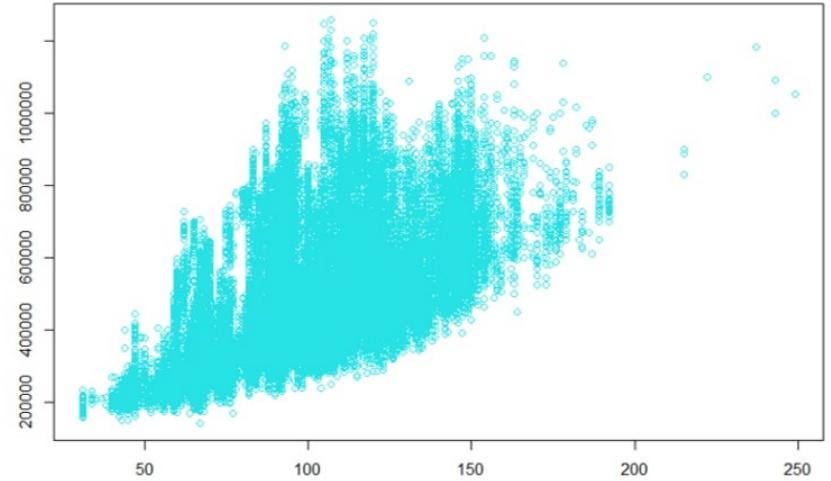
↑
Resale
Price

Positive Association

Two Quantitative Variables: Scatter Plots



Variability remains consistent as X increases



Variability increases stable as X increases

Interpreting a Scatter Plot: Key Questions

Is there an association?

Do points show a clear pattern, or do they appear random?

How strong is the relationship?

Do points lie close to the trend line, or are they widely scattered?

Positive or negative?

Positive: Y increases as X increases

Negative: Y decreases as X increases

Any unusual observations?

Do any points stand out from the overall pattern (outliers)?

Is it approximately linear?

Can the overall trend be reasonably represented by a straight line?

Is variability stable?

Does the spread of Y remain roughly constant as X changes, or does it fan out?

Correlation Coefficient



DEFINITION

The correlation coefficient measures the strength and direction of the linear relationship between two quantitative variables.

NOTATION

 r

FORMULA

Correlation coefficient between two quantitative variables X and Y:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{s_x} \right) \left(\frac{Y_i - \bar{Y}}{s_y} \right)$$

s_x, s_y : Standard deviation

$\bar{X}, \bar{Y}$: Mean

Properties of Correlation Coefficient

1

$$-1 \leq r \leq 1$$

2

Sign of r indicates direction:

- Positive r : positive association
- Negative r : negative association

3

Magnitude of r indicates strength:

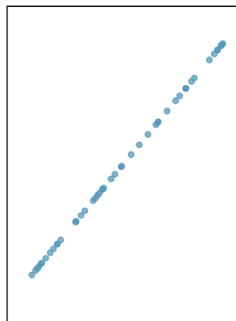
- The closer r is to 1, the stronger the positive linear association
- The closer r is to -1, the stronger the negative linear association.
- Values close to zero indicate a weak or no linear association.

4

Correlation between Y and X = Correlation between X and Y

Interpreting Correlation Values

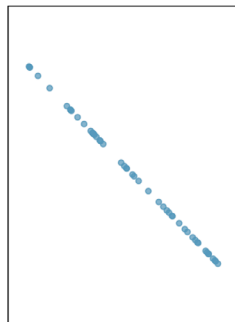
Perfectly Linear Relationship



$R = 1.00$

Perfect Positive

All points lie exactly on an upward straight line.

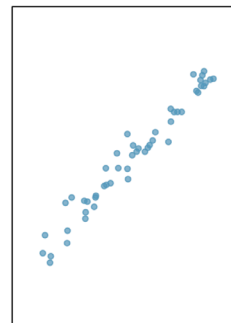


$R = -1.00$

Perfect Negative

All points lie exactly on a downward straight line.

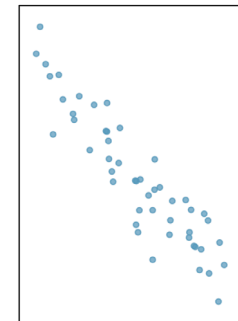
Strong Linear Relationship



$R = 0.98$

Strong Positive

Most points lie very close to a straight line, although not perfectly

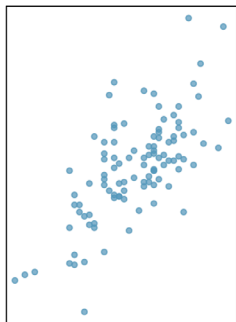


$R = -0.92$

Strong Negative

Interpreting Correlation Values

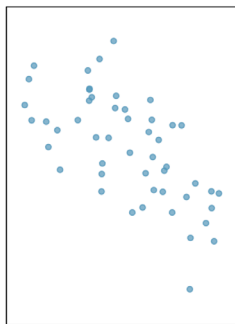
Moderate Linear Relationship



$R = 0.69$

Moderate Positive

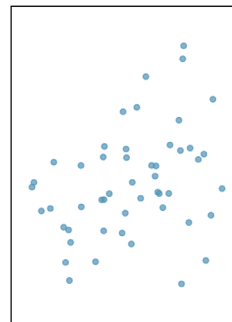
Points show a clear trend but with more variability around the line.



$R = -0.64$

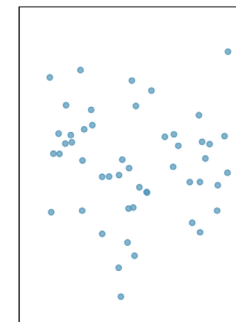
Moderate Negative

Weak Linear Relationship



$R = 0.33$

Weak Positive



$R = -0.08$

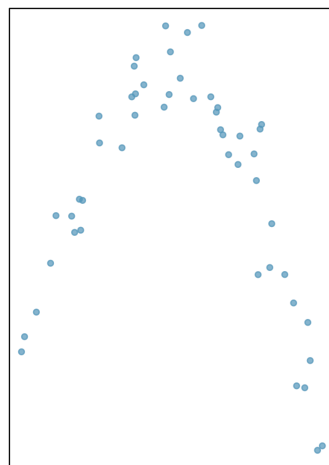
Weak Negative

The points appear largely random and do not form a clear increasing or decreasing linear trend.

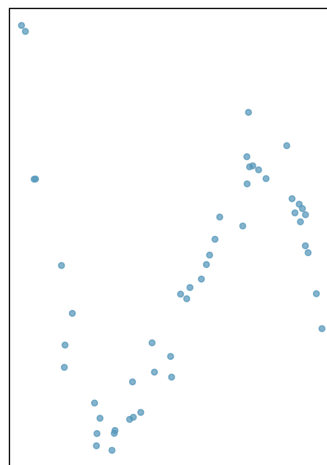
Limitations of the Correlation Coefficient

The correlation coefficient measures **LINEAR** association only.

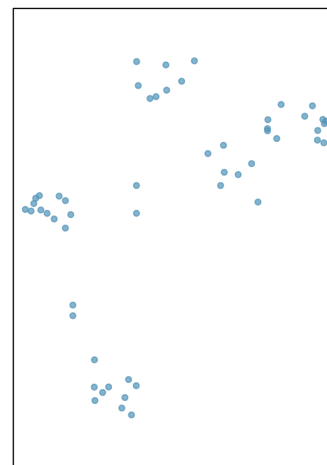
Non-linear relationships may produce misleadingly small r values:



$R = -0.23$



$R = 0.31$



$R = 0.50$

Limitations of the Correlation Coefficient



KEY INSIGHT

Correlation coefficient should never be *interpreted in isolation*.

Large r : does not guarantee the relationship is truly linear

Small r : does not necessarily mean no relationship exists

Exceptions: $r = -1$ or $r = 1$